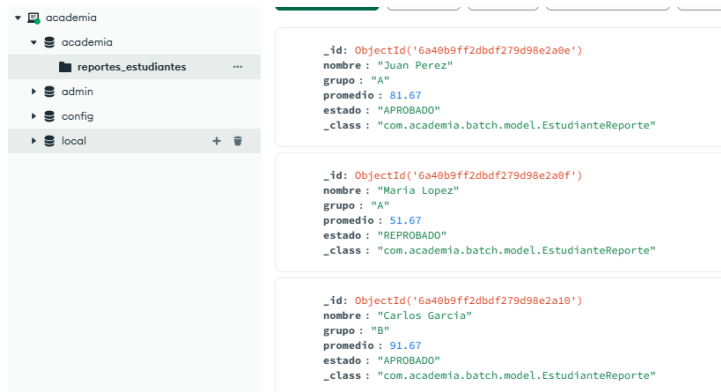
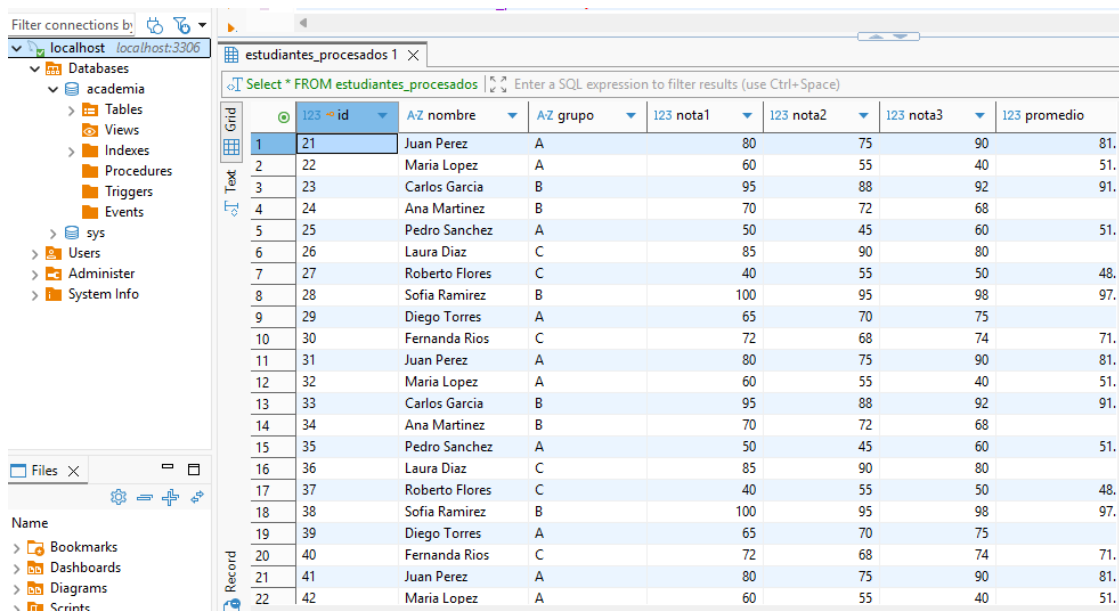


Evidencias del funcionamiento del proyecto

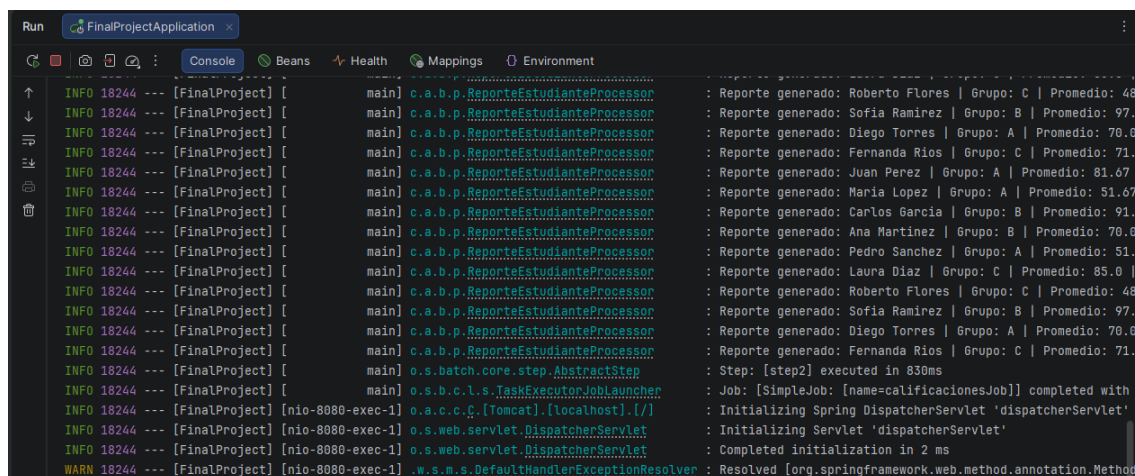
Creaciones de datos en MongoDB:



y MySQL



Y consola IntelliJ:



Pruebas con Curl:

curl <http://localhost:8080/api/estudiantes>:

Obtener todos los estudiantes:

```
C:\Users\monic>curl http://localhost:8080/api/estudiantes
[{"id":21,"nombre":"Juan Perez","grupo":"A","notal":80.0,"nota2":75.0,"nota3":90.0,"promedio":81.67}, {"id":22,"nombre":"Maria Lopez","grupo":"A","notal":60.0,"nota2":55.0,"nota3":40.0,"promedio":51.67}, {"id":23,"nombre":"Carlos Garcia","grupo":"B","notal":95.0,"nota2":88.0,"nota3":92.0,"promedio":91.67}, {"id":24,"nombre":"Ana Martinez","grupo":"B","notal":70.0,"nota2":72.0,"nota3":68.0,"promedio":70.0}, {"id":25,"nombre":"Pedro Sanchez","grupo":"A","notal":50.0,"nota2":45.0,"nota3":60.0,"promedio":51.67}, {"id":26,"nombre":"Laura Diaz","grupo":"C","notal":85.0,"nota2":90.0,"nota3":80.0,"promedio":85.0}, {"id":27,"nombre":"Roberto Flores","grupo":"C","notal":40.0,"nota2":55.0,"nota3":50.0,"promedio":48.33}, {"id":28,"nombre":"Sofia Ramirez","grupo":"B","notal":100.0,"nota2":95.0,"nota3":98.0,"promedio":97.67}, {"id":29,"nombre":"Diego Torres","grupo":"A","notal":65.0,"nota2":70.0,"nota3":75.0,"promedio":70.0}, {"id":30,"nombre":"Fernanda Rios","grupo":"C","notal":72.0,"nota2":68.0,"nota3":74.0,"promedio":71.33}, {"id":31,"nombre":"Juan Perez","grupo":"A","notal":80.0,"nota2":75.0,"nota3":90.0,"promedio":81.67}, {"id":32,"nombre":"Maria Lopez","grupo":"A","notal":60.0,"nota2":55.0,"nota3":40.0,"promedio":51.67}, {"id":33,"nombre":"Carlos Garcia","grupo":"B","notal":95.0,"nota2":88.0,"nota3":92.0,"promedio":91.67}, {"id":34,"nombre":"Ana Martinez","grupo":"B","notal":70.0,"nota2":72.0,"nota3":68.0,"promedio":70.0}, {"id":35,"nombre":"Pedro Sanchez","grupo":"A","notal":50.0,"nota2":45.0,"nota3":60.0,"promedio":51.67}, {"id":36,"nombre":"Laura Diaz","grupo":"C","notal":85.0,"nota2":90.0,"nota3":80.0,"promedio":85.0}, {"id":37,"nombre":"Roberto Flores","grupo":"C","notal":40.0,"nota2":55.0,"nota3":50.0,"promedio":48.33}, {"id":38,"nombre":"Sofia Ramirez","grupo":"B","notal":100.0,"nota2":95.0,"nota3":98.0,"promedio":97.67}, {"id":39,"nombre":"Diego Torres","grupo":"A","notal":65.0,"nota2":70.0,"nota3":75.0,"promedio":70.0}, {"id":40,"nombre":"Fernanda Rios","grupo":"C","notal":72.0,"nota2":68.0,"nota3":74.0,"promedio":71.33}, {"id":41,"nombre":"Juan Perez","grupo":"A","notal":80.0,"nota2":75.0,"nota3":90.0,"promedio":81.67}, {"id":42,"nombre":"Maria Lopez","grupo":"A","notal":60.0,"nota2":55.0,"nota3":40.0,"promedio":51.67}, {"id":43,"nombre":"Carlos Garcia","grupo":"B","notal":95.0,"nota2":88.0,"nota3":92.0,"promedio":91.67}, {"id":44,"nombre":"Ana Martinez","grupo":"B","notal":70.0,"nota2":72.0,"nota3":68.0,"promedio":70.0}, {"id":45,"nombre":"Pedro Sanchez","grupo":"A","notal":50.0,"nota2":45.0,"nota3":60.0,"promedio":51.67}, {"id":46,"nombre":"Laura Diaz","grupo":"C","notal":85.0,"nota2":90.0,"nota3":80.0,"promedio":85.0}, {"id":47,"nombre":"Roberto Flores","grupo":"C","notal":40.0,"nota2":55.0,"nota3":50.0,"promedio":48.33}, {"id":48,"nombre":"Sofia Ramirez","grupo":"B","notal":100.0,"nota2":95.0,"nota3":98.0,"promedio":97.67}, {"id":49,"nombre":"Diego Torres","grupo":"A","notal":65.0,"nota2":70.0,"nota3":75.0,"promedio":70.0}, {"id":50,"nombre":"Fernanda Rios","grupo":"C","notal":72.0,"nota2":68.0,"nota3":74.0,"promedio":71.33}]
C:\Users\monic>
```

Obtener uno por id existente:

```
C:\Users\monic>curl -i http://localhost:8080/api/estudiantes/25
HTTP/1.1 200
Content-Type: application/json
Transfer-Encoding: chunked
Date: Sun, 28 Jun 2026 06:54:26 GMT

{"id":25,"nombre":"Pedro Sanchez","grupo":"A","notal":50.0,"nota2":45.0,"nota3":60.0,"promedio":51.67}
```

Obtener uno por id inexistente:

```
C:\Users\monic>curl -i http://localhost:8080/api/estudiantes/2255
HTTP/1.1 404
Content-Length: 0
Date: Sun, 28 Jun 2026 06:55:48 GMT
```

crear registro:

```
C:\Users\monic>curl -i -X POST http://localhost:8080/api/estudiantes -H "Content-Type: application/json" -d '{"nombre":"Nuevo Alumno","grupo":"A","notal":80,"nota2":85,"nota3":90,"promedio":85}'
HTTP/1.1 201
Content-Type: application/json
Transfer-Encoding: chunked
Date: Sun, 28 Jun 2026 07:02:17 GMT

{"id":51,"nombre":"Nuevo Alumno","grupo":"A","notal":80.0,"nota2":85.0,"nota3":90.0,"promedio":85.0}
```

Nuevo Alumno	A	80	85	90	85
--------------	---	----	----	----	----

(PUT)reemplazar a

27	Roberto Flores	C	40	55	50	48.33
----	----------------	---	----	----	----	-------

Por:

```
C:\Users\monic>curl -i -X PUT http://localhost:8080/api/estudiantes/27 -H "Content-Type: application/json" -d '{"nombre":"Juan Perez","grupo":"A","notal":60,"nota2":90,"nota3":100,"promedio":83.33}'
HTTP/1.1 200
Content-Type: application/json
Transfer-Encoding: chunked
Date: Sun, 28 Jun 2026 07:15:30 GMT

{"id":27,"nombre":"Juan Perez","grupo":"A","notal":60.0,"nota2":90.0,"nota3":100.0,"promedio":83.33}
```

27	Juan Perez	A	60	90	100	83.33
----	------------	---	----	----	-----	-------

Cambio parcial: PATCH

14	34	Ana Martinez	B	70	72	68	70
----	----	--------------	---	----	----	----	----

A

```
C:\Users\monic>curl -i -X PATCH http://localhost:8080/api/estudiantes/34 -H "Content-Type: application/json" -d '{"grupo":"Z"}'
HTTP/1.1 200
Content-Type: application/json
Transfer-Encoding: chunked
Date: Sun, 28 Jun 2026 07:18:55 GMT
{"id":34,"nombre":"Ana Martinez","grupo":"Z","nota1":70.0,"nota2":72.0,"nota3":68.0,"promedio":70.0}
```

14	34	Ana Martinez	Z	70	72	68	70
----	----	--------------	---	----	----	----	----

Eliminar a:

5	25	Pedro Sanchez	A	50	45	60	51.67
---	----	---------------	---	----	----	----	-------

```
C:\Users\monic>curl -i -X DELETE http://localhost:8080/api/estudiantes/25
HTTP/1.1 204
Date: Sun, 28 Jun 2026 07:20:36 GMT
```

4	24	Ana Martinez	B	70	72	68	70
5	26	Laura Diaz	C	85	90	80	85

Obtener dato calculado por el servicio:

```
C:\Users\monic>curl http://localhost:8080/api/estudiantes/aprobados/total
{"aprobados":23}
```

Obtener reportes desde MongoDB:

[illegible]

Reprobados:

```
[{"id": "6a40b9ff2dbd72f9d98e2a0f", "nombre": "María Lopez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40b9ff2dbd72f9d98e2a12", "nombre": "Pedro Sanchez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40b9ff2dbd72f9d98e2a14", "nombre": "Roberto Flores", "grupo": "C", "promedio": 48.33, "estado": "REPROBADO", {"id": "6a40beb7269ca8ca6fea39df", "nombre": "María Lopez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40beb7269ca8ca6fea39e2", "nombre": "Pedro Sanchez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40beb7269ca8ca6fea39e4", "nombre": "Roberto Flores", "grupo": "C", "promedio": 48.33, "estado": "REPROBADO", {"id": "6a40beb7269ca8ca6fea39e9", "nombre": "María Lopez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40beb7269ca8ca6fea39ec", "nombre": "Pedro Sanchez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40bec0269e4a8ca6fea39ee", "nombre": "Roberto Flores", "grupo": "C", "promedio": 48.33, "estado": "REPROBADO", {"id": "6a40c0d43228a513c4e4db", "nombre": "María Lopez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40c0d43228a513c4e4de", "nombre": "Pedro Sanchez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40c0d43228a513c4e4dd", "nombre": "Roberto Flores", "grupo": "C", "promedio": 48.33, "estado": "REPROBADO", {"id": "6a40c0d43228a513c4e4dc5", "nombre": "María Lopez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40c0d43228a513c4e4dc8", "nombre": "Pedro Sanchez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40c0d43228a513c4e4dca", "nombre": "Roberto Flores", "grupo": "C", "promedio": 48.33, "estado": "REPROBADO", {"id": "6a40c0d43228a513c4e4dcf", "nombre": "María Lopez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40c0d43228a513c4e4dd2", "nombre": "Pedro Sanchez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO", {"id": "6a40c0d43228a513c4e4dd4", "nombre": "Roberto Flores", "grupo": "C", "promedio": 48.33, "estado": "REPROBADO"}}
```

Pruebas Unitarias:

```

Simbolo del sistema X + -
ficacionesJob]] completed with the following parameters: [{ 'run.id': '{value=7, type=class java.lang.Long, identifying=true}']] and the following status: [COMPLETED] in 2s947ms
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as a native agent to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
WARNING: A Java agent has been loaded dynamically (C:\Users\monic\m2repository\net\bytebuddy\byte-buddy-agent\1.17.8\byte-buddy-agent-1.17.8.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 17.26 s -- in com.academia.batch.FinalProjectApplicationTests
[INFO] Running com.academia.batch.processor.ProcessorTest
2026-06-28T01:46:12.983-06:00 INFO 8256 --- [FinalProject] [main] c.a.b.p.ReporteEstudianteProcessor : Reporte generado: null | Grupo: null | Promedio: 69.9 | Estado: REPROBADO
2026-06-28T01:46:13.007-06:00 INFO 8256 --- [FinalProject] [main] c.a.b.p.ReporteEstudianteProcessor : Reporte generado: null | Grupo: null | Promedio: 70.0 | Estado: APROBADO
2026-06-28T01:46:13.013-06:00 INFO 8256 --- [FinalProject] [main] c.a.batch.processor.EstudianteProcessor : Procesando estudiante: null | Grupo: null | Promedio: 90.0
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.041 s -- in com.academia.batch.processor.ProcessorTest
[INFO] Running com.academia.batch.service.EstudianteServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.754 s -- in com.academia.batch.service.EstudianteServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
=====
[INFO] BUILD SUCCESS
[INFO]
=====
[INFO] Total time: 33.042 s
[INFO] Finished at: 2026-06-28T01:46:14-06:00
[INFO]
=====

```